Grounding VI-SLAM noise injection in sensor physics

Literature synthesis · noise injection for visual-inertial SLAM

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The EuRoC sensor suite admits far fewer defensible corruptions than the robustness literature offers, and where it does admit them the physically correct magnitudes sit roughly an order of magnitude below the denoising community's defaults. The accepted camera noise model is the affine-variance Poisson–Gaussian form, stated normatively by EMVA 1288 as **$\sigma_y^2 = K^2\sigma_d^2 + \sigma_q^2 + K(\mu_y - \mu_{y.dark})$** (Jähne, EMVA 1288 R3.1 handout); carried through to 8-bit output for a well-exposed global-shutter machine-vision frame it yields **$\sigma \approx$ 0.5 grey levels in the dark and 1–2 grey levels at mid-grey**, so the σ = 5–20 DN additive-white-Gaussian default of the denoising literature corresponds to a 5–20× gained or underexposed sensor and must be declared as such rather than inherited silently. The MT9V034's **global shutter** and **mono8** output eliminate rolling shutter, demosaicing, chromatic and Bayer artifacts outright (ETH ASL EuRoC page), and the indoor Machine Hall venue eliminates the entire ImageNet-C weather family. On the inertial side, the four-parameter white-noise-plus-Wiener-bias model is *exactly* what ORB-SLAM3 assumes, so injecting it probes sensitivity to noise magnitude and not to model mis-structure; the measured insensitivity below ~10× is therefore unsurprising, because **OKVIS ships EuRoC white-noise densities 4–7× above the published values and VINS-Mono 24–40× above them** (OKVIS config; VINS-Mono config), placing the observed 20×/50× degradation knee just above the band the field already treats as reasonable. The measured **40–44 mm run-to-run ATE with a 3.4 mm pooled standard deviation is the resolution limit of the entire design**: the single 5× run that came in 6.2 mm better than clean is 1.8 pooled standard deviations from baseline and is not an improvement. Finally, the most complete SLAM perturbation benchmark published explicitly excludes IMU perturbations and explicitly declines to compose perturbations (Xu et al., arXiv:2402.08125), and no published study measures

trajectory error as a function of lens-contamination coverage — so this experiment occupies documented holes rather than inventing novelty, provided severities are expressed in physical units and trajectory validity is reported before ATE.

Global shutter and monochrome output disqualify a third of the standard corruption suite

The single most defensible move available in the experimental-design section is a hardware audit that *rejects* corruptions, because every rejection is a citation to the sensor rather than a taste judgment. The MT9V034 is a **1/3-inch, 752×480, 6.0 μm-pixel, global-shutter monochrome CMOS sensor with a 10-bit column-parallel ADC, >55 dB linear dynamic range and 4.8 V/lux-sec responsivity at 550 nm** (MT9V034 datasheet, ON Semiconductor Rev. 7, Jan 2017), and EuRoC runs two of them hardware-synchronized at 20 fps alongside an ADIS16448 at 200 Hz with Leica MS50 and Vicon ground truth (ETH ASL EuRoC page; Burri et al., IJRR 35(10):1157–1163, 2016). Rolling shutter is the clearest exclusion: the correct forward model requires re-rendering each row from a different pose, since **$t(f,r) = t_f + (R_o+r)/(R_o+R_h+R_1)\cdot T_{frame}$** (arXiv:1902.11084), which from an already-captured global-shutter frame can only be faked by a row-dependent warp with no occlusion handling. Conventional CMOS readout of 15–25 ms against EuRoC's 50 ms frame period would make rolling shutter a 30–50% intra-frame effect (readout-time compilation) — which is precisely why a global-shutter part was chosen for a MAV rig, and that design rationale is itself the justification sentence. If a rolling-shutter claim is wanted, the honest route is the TUM RS-VIO sequences accompanying Schubert et al., IROS 2019, not synthesis on EuRoC.

Three further exclusions follow from format and venue. Monochrome mono8 output removes chromatic aberration, colour-channel and Bayer/CFA demosaicing artifacts. The indoor Machine Hall removes fog, frost, snow, rain and lens flare — which together constitute the ImageNet-C "weather" family and the entire geometry-aware family of the closest SLAM corruption benchmark (Hendrycks & Dietterich, ICLR 2019; Thomas et al., arXiv:2608.30690), so the dominant corruption categories in the literature are inapplicable here and saying so pre-empts the obvious reviewer question. Dark current is a fourth quasi-exclusion on arithmetic grounds: measured at **0.283 e^-/s** on a modern comparable sensor at ~36 °C (Basler a2A2448-75um EMVA 1288 datasheet), integration over a 1–20 ms exposure yields 3×10^{-4} to 6×10^{-3} e^-, and even a 100× worse older part gives under 1 e^-, so dark current matters only as a fixed offset and as a hot-pixel mechanism, never as a temporal noise term at these exposure times.

Corruption	Admissible for this rig?	Governing evidence
Photon shot noise	**Yes, dominant** — Poisson, signal-dependent	EMVA 1288 photon transfer; Konnik & Welsh
Read noise (reset kTC + source follower)	**Yes** — Gaussian, signal-independent	Konnik & Welsh
Quantization	**Yes, but tiny** — σ = 0.29 DN for 1 LSB	Konnik & Welsh

Corruption	Admissible for this rig?	Governing evidence
Dark current / dark shot	**Offset and hot-pixel mechanism only** at 1–20 ms	Basler EMVA datasheet
Fixed-pattern noise (DSNU/PRNU, row/column)	**Yes, frozen per sensor** — must not be redrawn per frame	EMVA 1288 R3.1 decomposition
Row noise specifically	**Residual only** — MT9V034 has on-chip row-wise correction over 36 dark columns	MT9V034 datasheet
Dead / hot / stuck pixels	**Yes but rare** — 32-entry defect ROM, correction on by default	MT9V034 datasheet; Chapman et al.
Motion blur	**Yes, already present** in MH initialization motion	ETH ASL EuRoC page; RTAB-Map, arXiv:2403.06341
Illumination / exposure change	**Yes, already present and left/right asymmetric**	arXiv:2403.06341
Vignetting	**Physically yes, but the real V(x) is already baked into the raw frames**	Engel et al., arXiv:1607.02555; d'Angelo ICVS 2007
Lens occlusion (dust, droplets, mud)	**Mechanism yes, platform literature no** — all cited work is automotive	SoilingNet, arXiv:1905.01492; Porav et al., arXiv:1901.00893
JPEG / compression	**Not a capture artifact** (EuRoC ships lossless PNG); defensible as a telemetry artifact	Collaborative SLAM compression
Rolling shutter	**No — exclude.** MT9V034 is global shutter	ETH ASL EuRoC page; arXiv:2102.04036
Chromatic aberration, demosaicing, Bayer artifacts	**No** — monochrome mono8	MT9V034 datasheet
Fog, frost, snow, rain, flare	**No** — indoor Machine Hall	ETH ASL EuRoC page

One structural distinction matters more for a sequence-level SLAM study than it does for single-image denoising. Shot, dark-shot, read and quantization noise are redrawn every frame; **DSNU, PRNU, row/column offsets and defective pixels are frozen per sensor and must be drawn once and held fixed across the whole sequence**. Temporal noise is averaged away by multi-view triangulation and by the bundle adjustment; a frozen spatial pattern is not, and will instead bias every observation of every landmark that falls on it. Any injector that reseeds the FPN each frame is testing something physically impossible and will understate the damage. Sequence choice sharpens this further: MH_01_easy is an "easy" sequence with good illumination, rich features, low optical flow and essentially no motion blur, while the Machine Hall sequences deliberately open with fast motion for IMU initialization whose blur "is challenging for visual tracking" (ETH ASL EuRoC page) — so blur injection on MH_01 is adding to a near-zero baseline everywhere except the first few seconds, and that asymmetry should be reported rather than averaged over.

Affine variance pins realistic mono8 noise at 0.5–2 grey levels

The parametric model to cite has four independent statements of the same thing, which is as close to settled as this literature gets. EMVA 1288 gives it normatively as the photon transfer relation above, so **a = K (system gain, DN/e$^-$)** and **$b = K^2\sigma_d^2 + \sigma_q^2$**, and the two curves that determine the linear camera are sensitivity ($\mu_y - \mu_{y.dark} = K \cdot \eta \cdot \mu_p$) and photon transfer (Jähne handout). Learned raw denoising writes it as **$y \sim N(\mu = x, \sigma^2 = \lambda_{read} + \lambda_{shot} \cdot x)$** (Brooks et al., CVPR 2019); sRGB blind denoising as **$\sigma^2(L) = L \cdot \sigma_s^2 + \sigma_c^2$** (Guo et al., CVPR 2019); and the foundational raw-data treatment as Poissonian photon-sensing plus Gaussian stationary disturbance (Foi et al., IEEE TIP 17(10):1737–1754, 2008). Clipping at 0 and at saturation breaks the plain affine form and needs explicit handling (Foi, Signal Processing 89(12):2609–2629, 2009) — which matters for an 8-bit injector where clipping is routine. The Gaussian approximation to the Poisson term is safe here: electron counts at mid-grey are in the thousands, far above the >1000-photon threshold Konnik & Welsh give for Gaussian adequacy (arXiv:1412.4031).

The problem is that **no electron-level numbers are published for the MT9V034**. The Rev. 7 datasheet gives no full-well capacity, no conversion gain in μV/e$^-$ or DN/e$^-$, no read noise in electrons, no dark current, and no DSNU/PRNU specification, and no EMVA 1288 report for an MT9V034-based camera was located. Any electron-level claim about this sensor must therefore be bracketed from comparable parts or measured directly from dark and flat frames. The bracket runs as follows, and should be presented in the paper as derived arithmetic rather than as datasheet values. EMVA-style dynamic range is $20 \cdot \log_{10}(\mu_{e.sat}/\sigma_d)$, so ">55 dB linear" implies a saturation-to-read-noise ratio above 562; with a full well in the 15,000–33,000 e$^-$ range typical of its pixel-size class this gives **read noise of order 27–58 e$^-$**. If the 8-bit output spans the full well then 1 DN = FW/255 electrons, which for FW = 20,000 e$^-$ makes 1 DN = 78 e$^-$, and the noise budget at mid-grey becomes shot 100 e$^-$ = 1.28 DN, read 35 e$^-$ = 0.45 DN, quantization 0.29 DN, **total ≈ 1.4 DN**. The equivalent affine target is **$var(y) \approx 0.013 \cdot y + 0.28$ DN^2** at unity gain. Because EuRoC runs auto-gain, analog gain g scales $a \to g \cdot a$ and $b \to g^2 \cdot b_{read} + 1/12$, so at g = 4 the mid-grey sigma rises to roughly 3 DN; **$\sigma \approx$ 1–4 DN brackets plausible real EuRoC noise** without appealing to anything beyond the full-well assumption.

That bracket is the hinge of the whole camera arm, because it shows the denoising literature's defaults are not transferable. Brooks et al. sample **$\log \lambda_{shot} \sim U(\log 0.0001, \log 0.012)$** with **$\log \lambda_{read} \mid \log \lambda_{shot} \sim N(2.18 \cdot \log \lambda_{shot} + 1.2, 0.26)$**, tuned to the Darmstadt Noise Dataset (CVPR 2019); in 8-bit DN that λ_{shot} range corresponds to $a \in [0.026, 3.1]$ DN, which brackets the $a \approx 0.013$ DN derived above at its *low end* and extends about 200× upward to cover high-ISO consumer cameras. CBDNet samples **$\sigma_s \in [0, 0.16]$** and **$\sigma_c \in [0, 0.06]$** with JPEG quality in [60, 100] over 201 sampled camera response functions (Guo et al.) — and $\sigma_c = 0.06$ is about 15 DN of 8-bit read noise, deliberately far beyond a well-exposed industrial camera. The defensible position is to use the low end of these DND-calibrated ranges and say explicitly that the high end is a consumer-photography regime this sensor does not occupy. For an injector, the right pipeline is a *shortened* unprocessing in the sense of Brooks et al. — invert any companding (the MT9V034 can run a 12-to-10-bit companded mode), scale to electrons through an assumed full well, apply Poisson shot plus Gaussian read plus frozen PRNU/DSNU plus defects, then re-quantize to 8 bits with clipping — with Konnik & Welsh cited for the sensor-domain physics and Brooks et

al. for the invert-corrupt-reapply principle. A cheap, reviewer-friendly validation is to run Foi's released estimator or Liu, Tanaka & Okutomi's PCA-based noise-level estimator on real and synthetic frames and compare the recovered (a, b).

Spatial terms and defects are small but citable. Measured mono global-shutter cameras give **DSNU1288 of 0.3–1.9 e^- and PRNU1288 of 0.3–0.5%**, with the R4.0 breakdown showing DSNU1288.pix = 0.538 e^- against DSNU1288.col = 0.015 e^- and PRNU1288.col = 0.227% against PRNU1288.pix = 0.443% (Basler a2A2448; Basler acA1920-155um over 100 cameras; MATRIX VISION IMX487 R4.0 report). An older part like the MT9V034 plausibly sits at the 1–2% PRNU end that Konnik & Welsh call typical, with dark-current FPN factor 0.1–0.4 and column offset FPN around 0.10% (arXiv:1412.4031). For defects the field evidence is specific and limits the plausible severity sharply: the empirical in-field rate for CMOS APS pixels is **D = 10^(−1.13)·S^(−3.05)·ISO^0.505** defects per year per mm² (Chapman et al., Proc. SPIE 8659, 86590C, 2013), which at S = 6.0 μm and ISO 100 over the sensor's 12.99 mm² active area works out to about **0.04 new hot pixels per year** — roughly one per two decades. The MT9V034's 32-entry defect ROM bounds factory-mapped defects at **89 ppm**, correction is enabled by default, and bad pixels are replaced by neighbour averaging, so visible defects in EuRoC imagery are post-ROM residuals only. Two taxonomy facts are worth preserving because they change the model: the field studies found **no true stuck-high pixels at all**, concluding that apparent stuck-high behaviour is hot pixels with very large offsets, and some single-event upsets produce a **bright line** rather than a single bad pixel (Chapman et al.). A defensible defect injector therefore draws a frozen sparse mask at 1–100 ppm, assigns each defect an exposure-proportional dark rate R_dark with an optional constant offset b per I_offset = m(R_dark·T_e + b), and treats "stuck high" as the large-b limit rather than as its own class.

Blur, occlusion and compression need physical units, not ImageNet-C constants

The temptation with optical artifacts is to copy ImageNet-C's severity constants, and the single strongest technical reason not to is resolution. ImageNet-C's motion-blur ladder for images of side ≥ 65 px is **(radius, sigma) = (10,3), (15,5), (15,8), (15,12), (20,15)**, its defocus ladder **(3,0.1), (4,0.5), (6,0.5), (8,0.5), (10,0.5)**, and its JPEG ladder **Q = 25, 18, 15, 10, 7** (imagenet_c corruption source, third-party mirror, with the severity-3 motion-blur value independently cross-checked against Kamann & Rother's CVPR 2020 supplemental). A radius-15 kernel is 6.7% of image width at 224 px but only 2% at 752 px, so the same nominal severity means a materially different physical perturbation — and the ImageNet-C authors themselves warn that 32×32 corruption robustness "is not predictive of corruption robustness on larger images" (ar5iv 1807.01697). The fix is to parameterize blur in **pixels of image-plane displacement**, which is also what connects it to the hardware: the displacement of a static point is approximately **d ≈ f_px·ω·t_exp + f_px·v_perp·t_exp/Z** (derived from the pinhole model, not a quoted result), so rotation blur is depth-independent and dominates at normal depths while translation blur is depth-dependent and therefore spatially varying, which a single global convolution kernel cannot represent.

EuRoC makes the physically faithful synthesis available for free. The literature's formation model is the time integral over the exposure, $\mathbf{B(x) = \lambda \int_0^{\tau} I_t(x)\, dt \approx (1/n) \Sigma_i I_i(x)}$, with the within-exposure trajectory parameterized by two poses interpolated in the Lie algebra as $\mathbf{T_t = T_{start} \cdot exp((t/\tau) \cdot log(T_{start}^{-1} T_{end}))}$ (Liu et al., MBA-VO, ICCV 2021) — and MBA-VO's own real dataset is 752×480 at 27 fps with 100 Hz mocap, essentially this imaging geometry. With 200 Hz IMU and 20–100 Hz ground-truth pose in hand, the right injector interpolates the ground-truth pose over an assumed exposure window, warps the frame along that path, and averages N samples; a uniform linear kernel remains defensible as the cheap ablation, labelled as the depth-independent approximation. GoPro and REDS both derive blur by frame-averaging high-frame-rate video (Nah et al., CVPR 2017, 3,214 images at 1280×720; Nah et al., NTIRE 2019, 30,000 frames), so frame-averaging is also the approach most consistent with the dominant literature. The one missing number is the exposure time: **no published figure for EuRoC's actual exposure times was found**, and the closest comparable global-shutter machine-vision rig is the TUM mono benchmark at **0.05–20 ms** (Engel et al., arXiv:1607.02555). Read it from the bags if present, or state 1–20 ms as an assumption with the TUM figure as precedent.

Blur also carries the clearest evidence of real damage and the clearest internal conflict. ORB-SLAM dropped **5–46% of frames** on motion-blurred TUM RGB-D sequences against 0% for MBA-VO, and on blurred ArchVizInterior sequences MBA-VO reached ATE 0.0256/0.0184/0.0202 m against DSO's 0.2132/0.1655/0.1286 m and ORB-SLAM's 0.0325/0.0122/0.1008 m (MBA-VO). Feature-level numbers show the same direction: under Blur-HPatches at increasing severity, SIFT repeatability falls 56.99% → 45.94% and SuperPoint 58.60% → 43.28%, with matching accuracy at 3 px dropping to about 25% for the strongest baseline (Zhao et al., SSMB, arXiv:2608.27181). ORB specifically is reported to fail faster than scale-space detectors — "ORB features are missing at higher blur levels and their position is less stable" (Aldana-Iuit et al., ICPR 2016) — and KAZE/AKAZE win on blurred imagery (arXiv:2012.04135). **This conflicts with a broad empirical evaluation reporting that the FAST/ORB combination had the fewest failed matches (100) under blurring conditions** (PMC11623105); the discrepancy plausibly reflects different blur magnitudes and different metrics (failed-match count versus repeatability) and should be reported as an unresolved conflict rather than silently resolved. The mechanism argues for measuring count and repeatability separately: FAST thresholds on an absolute intensity difference over a contiguous ring arc, so low-pass filtering pushes candidates below threshold in a cliff rather than a slope, and ORB's Harris re-ranking and intensity-centroid orientation are both contrast-dependent, so surviving keypoints also get worse orientations. Critically, **ORB-SLAM3's extractor uses a grid plus adaptive-threshold fallback to hit a keypoint budget**, which means count-based metrics on the full system will understate damage; inlier ratio and repeatability are the informative quantities.

Photometric corruptions are best framed as negative controls for a feature-based system. The standard model is the affine brightness transfer $\mathbf{I' = clip(a \cdot I + b)}$, and the DSO/TUM photometric model is $\mathbf{I(x) = G(t \cdot V(x) \cdot B(x))}$ with the vignetting map normalized to $\max(V) = 1$ (Engel et al.). Vignetting itself is modelled either by the $\cos^4$ closed form $\mathbf{V(r) = 1/(1 + (r/f)^2)^2}$ or, because $\cos^4$ "does not model real cameras and lenses well," by an even radial polynomial $V(r) = 1 + \Sigma \beta_n r^{2n}$ (d'Angelo, ICVS 2007), with machine-vision lenses typically specifying **45–52% corner relative illumination** and ~40% as the practical floor (Edmund Optics imaging catalogue p.21). A sweep over corner attenuation {0.9, 0.8, 0.65,

0.5, 0.4} therefore spans "better than typical" to "worse than a typical lens." Two caveats are load-bearing. First, **EuRoC ships geometric but not photometric calibration, so the rig's true vignetting is already present and uncorrected in the distributed frames**; synthetic vignetting stacks a fictitious falloff on a real one, and that is a stated limitation, not a detail. Second, a smooth multiplicative gain field nearly constant across a 31×31 patch largely cancels in FAST's centre-versus-ring comparison and in BRIEF's pairwise comparisons, so the expected effect on ORB is small while the effect on direct methods is large — consistent with the reported asymmetry that photometric calibration improves DSO significantly while *reducing* ORB-SLAM's overall performance (Yang, Wang & Cremers, IEEE RA-L 2018, arXiv:1705.04300; **this specific sentence came from a search-result summary, not verified full text — see the caveats section**). The informative photometric regime for a feature-based front end is therefore temporally *varying* gain and offset, where clipping destroys gradient information irreversibly, and EuRoC's documented left/right exposure asymmetry — "exposure level between the cameras is not [synchronized] (e.g. right image can be darker with lower contrast than the left one)" (arXiv:2403.06341) — makes an *asymmetric, single-camera* illumination corruption the most physically faithful variant for this stereo rig.

Lens occlusion is the arm with the strongest mechanism and the weakest platform precedent, and the report should say both. The entire soiling literature is automotive — Valeo surround-view fisheye and windshield deraining — and **no work on lens contamination for indoor or MAV platforms was found**, so transfer must be argued on mechanism rather than on cited indoor results. What transfers cleanly is the taxonomy, the formation model and the severity unit. Valeo's taxonomy is by optical transmission, not material: two classes (opaque, transparent) in SoilingNet, extended to four (clean, transparent, semi-transparent, opaque) in TiledSoilingNet, with the 2025 re-benchmark restating them precisely (Beránek et al., arXiv:2511.09740). The compositing model is $\mathbf{I = (1 - M) \odot B + R}$ (Qian et al., CVPR 2018), and the field's native severity unit is TiledSoilingNet's **coverage** — per-class area fraction per tile over a 4×4 grid, with class coverages summing to one — so an injector reporting per-frame or per-tile coverage fraction is speaking the literature's own language. What does *not* transfer is geometry: von Bernuth's mm-scale droplet radii are physical sizes on a windshield **30 cm from the lens at 27° tilt**, so the mm-to-pixel mapping changes with the rig and coverage fraction in pixels is the portable parameter. A MAV has no windshield, no wiper and no road spray, but does encounter dust, fingerprints on a lens cover and condensation — which is exactly the failure list QueensCAMP emulates on real indoor RGB-D data (lens dirt, condensation, underexposure, overexposure), making it the best available platform-adjacent precedent.

The occlusion parameterization has a documented three-tier fidelity ladder, which lets a cheap injector cite the expensive one as its reference. At the top, physically correct ray-traced rendering models each drop as a **sphere cap with $h = \tan(\theta/2) \cdot d$ at a contact angle of about 87° for ~2 mm radii**, refracts per Snell's law at entry and exit with total internal reflection, samples scene colour from a reconstructed point cloud, convolves with a disk kernel for defocus, and draws radii from $\mathbf{r \sim N(1.5\ mm,\ 0.4\ mm)}$ — at 30.5 dB PSNR for a drop-free re-render and about 3 minutes per KITTI image (von Bernuth, Volk & Bringmann, IEEE IV 2018). In the middle, Porav et al.'s shader approach encodes refraction offsets in the red and green channels of a proto-droplet texture and thickness in blue, giving $\mathbf{x_r = u + (R \cdot B)}$, masks it with an alpha layer, seeds droplets with per-pixel probability $\mathbf{P_r}$ scaled anisotropically by $\mathbf{S_x}$, $\mathbf{S_y}$, merges

neighbours with metaballs, and gates streaking on a **4 mm diameter threshold with slip drawn from N(0,3)** (ICRA 2019, arXiv:1901.00893). At the bottom, plain alpha compositing follows Eigen/Qian with a mask and an opacity. The minimal defensible parameter set across all three is coverage fraction, opacity, blob-size distribution, blur radius and spatial correlation, and every one of those has a direct citation. For severity anchoring, the only hard coverage numbers available are class balances from curated soiling subsets, not deployment distributions: the WoodScape soiling subset's **5,000 annotated images are 39% clear, 12% transparent, 19% semi-transparent and 30% opaque by pixel** — about 61% soiled area with 30% fully opaque (arXiv:2511.09740) — and SoilingNet's internal training set has **52% of frames carrying some soiling** (arXiv:1905.01492). **No fleet-level coverage distribution and no cleaning-trigger threshold exists in public sources**; only cleaning-hardware patents were found. Two mechanistic findings should shape expectations. Detection accuracy is systematically worst on the *transparent* class (58% in TiledSoilingNet), so low-alpha occlusion is both hardest to detect and most likely to corrupt estimation silently. And the motion observed inside adherent droplets is **1/20 to 1/30 slower than the scene** (Porav et al.) — meaning occluded regions do not merely lose features, they emit systematically wrong, spatially coherent apparent motion, which is a worse failure mode for an optimizer than outright feature loss because it generates consistent-looking but biased correspondences. That argues the dominant soiling failure mode for ORB-SLAM3 is drift and bias, not track loss.

Compression closes the list. JPEG is fictitious as a capture artifact for lossless-PNG EuRoC, but it is a genuine deployment artifact for a MAV compressing imagery before a radio downlink to an offboard SLAM process, and that bandwidth framing is the one the collaborative and edge-SLAM compression literature also uses (TUM collaborative SLAM compression; arXiv:2603.11085). Its artifact structure is adversarial specifically for corner detectors: 8×8 DCT block boundaries create spurious high-contrast corners on a regular grid while high-frequency quantization erodes true corners, so the expected failure is displaced and spurious keypoints rather than simply fewer of them, and repeatability should degrade faster than count. Detector-level evidence exists but not for ORB: SURF holds to ~95% compression ratio and Harris-Laplace and Hessian-Laplace stay high-repeatability under heavy JPEG, while SIFT is reported usable only to ~50% and can collapse near-zero at 98% (Ehsan et al., arXiv:1605.05791; **search-summary only, and expressed in compression ratio rather than quality factor — do not convert**). At the localization level modern codecs are surprisingly benign, with JPEG-XL at quality 30 reported not to decrease localization performance significantly (arXiv:2601.04185; **search summary**). A sweep over **Q ∈ {90, 70, 50, 30, 15, 7}** captures the knee while anchoring the low end to ImageNet-C's Q = 25…7 for cross-paper comparability.

Production IMU configs already sit 4–40× above the identified densities

The inertial side is the cleanest part of the design because the model is unambiguous and the units are physical. Every modern VI-SLAM system uses the same minimal two-term stochastic model per axis — additive continuous-time white Gaussian noise plus a bias modelled as a Wiener random walk, $\tilde{\omega}(t) = \omega(t) + b_g(t) + \eta_g(t)$ with $\dot{b}_g(t) = \eta_{bg}(t)$ (Kalibr IMU Noise Model) — parameterized by four continuous-time densities with units rad/s/√Hz, rad/s²/√Hz, m/s²/√Hz and m/s³/√Hz. The identification method is Allan deviation on hours of stationary data, with the **−1/2 log-log slope giving the white-**

noise density read at $\tau = 1$ s and the **+1/2 slope giving the bias random walk**, and bias instability extracted as the minimum ADEV divided by **0.664** per the IEEE Std 952 convention (InertialSense Allan Deviation; visual-slam-roadmap); the canonical reference is El-Sheimy, Hou & Niu, IEEE T-IM 57(1):140–149, 2008, and practical tooling requires at least 3 hours of stationary data (allan_variance_ros). Note the structural consequence: **bias instability (the flat, flicker-noise region) has no slot in the four-parameter VI-SLAM model**, so flicker noise is simply not representable, and practitioners paper over it by inflating the random-walk term.

The discretization is where injectors break, and it is worth stating verbatim in the paper. For white noise the per-sample standard deviation is the density **divided** by $\sqrt{\Delta t}$, $\sigma_d = \sigma_c \cdot \sqrt{f_s}$; for the bias random walk the per-step increment is the diffusion density **multiplied** by $\sqrt{\Delta t}$, $b_k = b_{k-1} + \sigma_b \cdot \sqrt{\Delta t} \cdot w_k$ (Kalibr wiki). The two scale in opposite directions in Δt because white noise is a measurement perturbation spread over a wider Nyquist band while the bias random walk is a state perturbation accumulated over time, and at 200 Hz the factor $\sqrt{f_s} = 14.14$ means an injector that applies $\sqrt{\Delta t}$ where $\sqrt{f_s}$ belongs is off by $f_s = 200$. This is not a hypothetical: `gaowenliang/imu_utils` reports **N·√f rather than N** for white noise, and a bug of exactly this form was filed against GTSAM's `CombinedImuFactor` (ROS Discourse; GTSAM issue 252). The OKVIS EuRoC config itself carries a units typo, commenting `sigma_gw_c` as "[rad/s^s/sqrt(Hz)]" and `sigma_aw_c` as "[m/s^2/sqrt(Hz)]" when they should be rad/s²/√Hz and m/s³/√Hz (OKVIS config). The clean parameterization for the experiment is therefore to hold the four continuous densities as the knobs and apply a multiplier k inside the injector, which keeps results rate-independent and directly comparable to published configs.

Source	σ_g [rad/s/√Hz]	σ_bg [rad/s²/√Hz]	σ_a [m/s²/√Hz]	σ_ba [m/s³/√Hz]
EuRoC published (ADIS16448, 200 Hz)	1.6968e−04	1.9393e−05	2.0000e−03	3.0000e−03
ORB-SLAM3 EuRoC.yaml	1.7e−04 (1.0×)	1.9393e−05 (1.0×)	2.0000e−03 (1.0×)	3.0000e−03 (1.0×)
OpenVINS EuRoC chain	1.6968e−04 (1.0×)	1.9393e−05 (1.0×)	2.0000e−03 (1.0×)	3.0000e−03 (1.0×)
OKVIS EuRoC	1.20e−03 (**7.1×**)	4.0e−06 (0.21×)	8.0e−03 (**4.0×**)	4.0e−05 (**0.013×**)
VINS-Mono EuRoC	4.0e−03 (**23.6×**)	2.0e−06 (0.10×)	8.0e−02 (**40×**)	4.0e−05 (**0.013×**)
ADIS16448 datasheet typ. (converted)	2.356e−04 (from 0.0135 °/s/√Hz)	—	not verified	not verified
TUM-VI (Bosch BMI160, Allan-derived)	8.0e−05	2.2e−06	1.4e−03	8.6e−05

Sources: OpenVINS kalibr_imu_chain.yaml; ORB-SLAM3 EuRoC.yaml; OKVIS config; VINS-Mono config; ADIS16448 parametric table; TUM-VI, arXiv:1804.06120.

Two readings of that table matter for the experiment. First, the practice is not uniform inflation but **re-balancing**: every system that departs from the EuRoC values moves white noise up by 4–40× and bias random walk down by 5–75×. Both OKVIS and VINS-Mono land on an accelerometer random walk about **75× below** the EuRoC-published 3.0e−3, and TUM-VI's Allan-derived BMI160 figure is 35× lower still, which together suggest the EuRoC-published accelerometer random walk is itself too large — so a multiplier sweep anchored at k = 1 on that parameter starts from an already-inflated baseline and should say so. The mechanism is tuning rather than physics: inflating white noise down-weights the IMU relative to vision in the information balance, while deflating bias random walk keeps the bias estimate from wandering into real motion. Second, this band is the citable justification for the sweep range. **Because production configs already sit 4–40× from the identified white-noise density, a sweep over multipliers of roughly 1× to 40× is squarely inside what the field treats as reasonable, and multipliers beyond ~40× are the genuinely adversarial regime.** Kalibr's own documentation supplies the reason inflation is legitimate at all: the noise model "is derived from a sensor which does not undergo motion, and at constant temperature," and therefore represents an **optimistic** assessment (Kalibr wiki).

This is exactly where the measured sweep lands, and the agreement is strong enough to carry a results paragraph. ORB-SLAM3 was essentially insensitive to gyro/accel white noise below about 10× the sensor's own density, degraded by **+26 mm at 20×** and by **+209 mm at 50×**. Against a 3.4 mm pooled standard deviation those are roughly **7.6σ and 61σ** respectively, so both are unambiguously real, while the single 5× run that came in **6.2 mm better than clean is 1.8σ** and should be reported as indistinguishable from baseline rather than as an improvement. The no-effect region below 10× is precisely the band OKVIS and VINS-Mono already occupy without breaking, which turns an apparently null result into a positive, citable finding: **ORB-SLAM3's information balance tolerates the full range of white-noise inflation the field uses in practice, and the knee lies just above it.** Two further observations sharpen the interpretation. The ADIS16448 is best described not as low-noise but as *well-calibrated* — its raw gyro noise density of 0.0135 °/s/√Hz is essentially equivalent to a Bosch BMI055 consumer part at 0.014 dps/√Hz (Bosch Sensortec forum), and what distinguishes it is bias specification (14.5 °/hr in-run, 0.05° axis alignment, factory temperature calibration over −40 to +85 °C). So inflating white noise toward consumer levels is only a 2–3× perturbation, whereas inflating bias random walk toward uncalibrated-part levels is a far larger one — which is a direct argument that **the bias random walk, not white noise, is the under-explored axis**, and no published study sweeps it independently. And the asymmetry to expect is accelerometer-dominant, because accelerometer error integrates twice into position while gyro error integrates once into attitude, and visual updates constrain orientation more tightly than they constrain a gravity-aligned acceleration bias.

The deterministic error terms are the other half of the story and the place where this specific IMU licenses a corruption the VI-SLAM model cannot absorb. The full textbook model is $\tilde{\omega} = (I + S_g + M_g)\omega + G_g \cdot a + b_g(t) + \eta_g(t)$, and the standard VI-SLAM reduction sets S = M = G = 0 and ignores temperature. For the ADIS16448 the datasheet quantifies exactly those dropped terms: **axis-to-axis alignment 0.05°, linear-g sensitivity 0.015 °/s/g, bias repeatability 0.5 °/s, in-run gyro bias stability 14.5 °/hr, accelerometer in-run bias stability 250 μg** (ADIS16448 parametric table). Injecting white noise and bias random walk tests the estimator against *its own assumed model*, which isolates magnitude

sensitivity and is honestly a lower bound on real degradation; injecting scale factor, misalignment or g-sensitivity tests structural mismatch, which is the strictly harder and more realistic regime — and the magnitudes are datasheet-justified for this part rather than invented. Kalibr's extended multi-IMU per-axis calibration and OpenVINS's `rpng` model restore S and M explicitly (Rehder et al., ICRA 2016; OpenVINS chain config), so the model class is established even though no robustness study sweeps it.

Severity calibration has five rungs, and physical units beat inherited constants

The most useful thing the corruption-benchmark literature offers is not its parameters but a defensibility ordering for how parameters are chosen, and it can be stated as five published strategies in ascending order. The weakest is to inherit ImageNet-C's constants — which is what Robust-SLAM does for images ("we follow the severity used for the most prevalent image robustness benchmark") and ModelNet40-C does wholesale ("created with five severity levels for each corruption type, the same as ImageNet-C") (ar5iv 2402.08125; ar5iv 2201.12296). This is weak because **ImageNet-C documents no calibration procedure and no parameter ranges in the paper at all**; the severities exist only as constants in released code, and two independent reads of both the v1 preprint and the ICLR version found the five-level structure asserted with no derivation (ar5iv 1807.01697; ar5iv 1903.12261). Second is discrete configuration enumeration where no continuous axis exists — ModelNet40-C substitutes **five view angles** for five severities when LiDAR and occlusion severity proved unquantifiable, preserving the reporting structure. Third is deriving parameters from a real reference dataset's empirical distribution, as Robust-SLAM does for depth ("we refer to the depth range and distribution in real-world dataset TUM-RGBD"), giving a clipping range of **0.42–10 m**. Fourth is physical units from a physical model: Dong et al. tie snowfall to meteorological rates of **{0.20, 0.73, 1.5625, 3.125, 7.29} mm/hr** and fog to attenuation $\alpha \in \{0.005, 0.01, 0.02, 0.03, 0.06\}$ (ar5iv 2303.11040), and Robo3D publishes per-dataset parameter tables where the same nominal severity maps to **different** physical parameters on different sensors — motion blur σ_t of {0.20, 0.25, 0.30} on SemanticKITTI-C but {0.04, 0.08, 0.10} on KITTI-C (ar5iv 2303.17597). Fifth and strongest is interpretable real-world units plus an automated search for the failure threshold, which is what SLAM Adversarial Lab does with "meters for fog visibility" and "an automated search procedure for identifying failure thresholds" (Hefny, Dantu & Ko, IROS 2026, arXiv:2603.17165).

For this experiment rungs four and five are both available, which is the design's strongest claim. The IMU densities in rad/s/√Hz and m/s²/√Hz *are* physical units identified by Allan variance from the sensor itself; the camera's affine coefficients in DN and DN² are physical units derived from EMVA 1288; blur displacement in pixels and coverage fraction are dimensionless or physical; and the measured 20×/50× knee already constitutes an empirical failure-threshold search on the inertial axis. Four more conventions are worth adopting verbatim because each is precedented. **Geometric rather than linear spacing** of multipliers is better supported: Robust-SLAM uses 2×/4×/8× motion speed-up, Dong et al.'s snowfall rates are approximately geometric, and the 2.512× figure attributed to the subterranean VIO benchmark is 10^0.4, implying a log-spaced sweep in 4 dB steps (ar5iv 2402.08125; arXiv:2609.18628 — **the 2.512× figure is unverified; see caveats**). **Three levels if runs are expensive, five if not**, with the

reason stated: the two robotics benchmarks that evaluate many expensive models use three (Robo3D with 34 models, RoboBEV with 33), and the one treating corruption count as the contribution uses five (Dong et al., 27 types × 5 = 135). **Randomize nuisance parameters within a level** — ImageNet-C's explicit design choice ("Real-world corruptions also have variation even at a fixed intensity... each fog cloud is unique to each image") and Robo3D's uniform sampling of α — and report mean with min/max across repeats. And **hold severity constant per sequence** (Robust-SLAM's static mode) when severity must be a clean independent variable, varying it frame-to-frame only when realism is worth confounding attribution for.

The claim that must *not* be asserted is the folk one. **No benchmark in this literature calibrates severity against a measured perceptual or task-destruction criterion**, and no psychophysical validation of the "barely visible to task-destroying" span exists in any paper examined. The defensible move is to demonstrate the span post hoc from the results — show that the lowest level produces ATE statistically indistinguishable from clean and the top level produces divergence or track loss in a stated fraction of runs — with SLAM Adversarial Lab's failure-threshold search as the methodological precedent for making the top of the range *defined by* task destruction rather than asserted to be there. Mintun et al. supply one further concrete safeguard: because corruption robustness degrades significantly when test corruptions are **perceptually dissimilar** from the training corruption family, and they quantify this with a Minimal Sample Distance measure and a deliberately dissimilar CIFAR-10-C̄ control set (NeurIPS 2021, arXiv:2102.11273), including at least one deliberately dissimilar corruption arm — structured or heavy-tailed rather than Gaussian — demonstrates that observed robustness is not an artifact of one corruption family.

Report validity before ATE, and the 3.4 mm floor before both

The metric conventions in this literature converge on one rule with two layers. The rule is that **a continuous error metric alone is not reportable**, because a diverged or track-lost run yields a finite but meaningless number that makes catastrophic failure look mild — or, worse, look better than graceful drift. Robust-SLAM's answer is the triple of **ATE, Success Rate from cumulative consecutive pose distances, and Cumulative Success Rate as the percentage of sequences within an ATE threshold**, reported as mean with max/min across perturbation settings averaged over multiple runs (ar5iv 2402.08125). Failure-or-Drift's answer is to report **Trajectory Validity V_r as a binary first**, then APE_RMSE and RPE_RMSE conditioned on a valid output (arXiv:2608.30690), and its numbers show why: **ORB-SLAM2 returned valid trajectories in only 37 of 63 image-space, 15 of 23 geometry-aware and 10 of 15 compound attempts**, and "Gaussian noise, impulse noise, and snow produce no evaluable ORB-SLAM2 trajectory at any recorded severity," while DPVO and DROID-SLAM remained valid in every synthetic attempt and merely drifted. That is a strong a-priori prediction for an ORB-SLAM3 experiment: **expect loud failure, not graceful degradation, as the dominant mode for a feature-based front end**, and design the reporting so a run that produces nothing is distinguishable from a run that produces garbage.

The same paper supplies two warnings that should shape the results tables. Severity does not reliably order estimation difficulty — under geometry-aware rain, **DROID-SLAM's APE decreased from 111.9 m at severity 1 to 87.9 m at severity 5** — so the full severity curve must be reported rather than any single

aggregate robustness number, and the authors explicitly caution that "severity orders the corruption" but not the difficulty. And method rankings flip with corruption *type*: DPVO had lower APE in 30 of 43 paired image-space settings while DROID-SLAM had lower APE in **33 of 33** geometry-aware settings, with severity-curve correlations ranging from $\rho = 1.00$ (low light versus brightness+contrast for DPVO) to $\rho = -0.10$ (rain versus rain+contrast). The practical consequence is that **an occlusion-type corruption is not substitutable by Gaussian noise or blur**, which is itself the argument that adding a soiling arm to a noise/blur suite is a substantive addition rather than a redundancy; and that optical/temporal artifacts should be kept separate from sensor-noise artifacts in both design and reporting rather than pooled into one robustness score. For aggregation, the normalization idea transfers cleanly: ImageNet-C's **CE_c = Σ_s E_{s,c} / Σ_s E_{s,c}^AlexNet** with mCE as the average, plus a clean-subtracted relative mCE, and Robo3D's parallel pair of **mCE against a fixed reference model** (CenterPoint for detection, MinkUNet for segmentation) and **mean Resilience Rate RR_i = Σ_l Acc_{i,l}/(3·Acc_clean)** measuring performance retained relative to the model's *own* clean score (ar5iv 1807.01697; ar5iv 2303.17597). **No SLAM-specific normalized aggregate analogous to mCE was found**, so defining one — a clean-normalized mean corruption ATE with a resilience-rate companion — is a contribution rather than a convention to cite. ATE and RPE themselves trace to Sturm et al., IROS 2012 and are conventionally computed with `evo`. A more ambitious option exists if the trajectory-level arm needs it: SubT-MRS defines **position and rotation robustness R_p and R_r as the area under F-1-score curves over velocity estimates**, on the argument that ATE is a poor proxy for the quantity a downstream controller cares about (arXiv:2307.07607).

What the literature does *not* supply, and what the measured nondeterminism floor does, is the resolution limit of the whole design. Run-to-run ATE on identical input is **40–44 mm with a pooled standard deviation of 3.4 mm**, an ~8% coefficient of variation, and every injected effect must be read against it. The derived consequence is a minimum detectable effect: at 80% power and two-sided $\alpha = 0.05$, $\text{MDE} = 2.80 \cdot \sigma \cdot \sqrt{(2/n)} \approx 9.52 \cdot \sqrt{(2/n)}$ mm, which is **6.0 mm with 5 runs per arm, 4.3 mm with 10, and 3.0 mm with 20** (derived arithmetic, not a cited result). Two implications follow directly. The 6.2 mm "improvement" at 5× IMU white noise is right at the 5-run detection threshold and was observed in a single run, so it is noise; and **any claimed denoising benefit smaller than roughly 10–15% of baseline ATE is unmeasurable without on the order of ten or more seeds per condition**. This cuts both ways and should be stated as a design constraint rather than buried: it bounds what the denoise-before-SLAM and denoise-after-SLAM arms can possibly demonstrate, and it means the seed count per condition is a first-class experimental parameter, not an afterthought. Reporting the clean-input run-to-run distribution as an explicit control band on every figure is the honest presentation, and it is precedented in spirit by Robust-SLAM's practice of reporting mean with max/min over repeated runs.

For the denoise-after-SLAM arm, trajectory-space perturbation has its own separate precedent worth citing: Robust-SLAM treats **rotation deviation σ = 1°, 3°, 5° and translation deviation σ = 0.0125, 0.025, 0.05 m** as first-class corruption types (ar5iv 2402.08125), and the moment-matching VI-SLAM work injects **σ = 0.1 rad rotation noise on roll, pitch and yaw** to simulate poor-quality pose estimates (arXiv:2205.13821). Those are the natural anchors for the output-trajectory arm, and they are usefully at a different scale from each other — 5° versus 0.1 rad ≈ 5.7° happens to agree, which is a convenient cross-check.

Four literature holes this experiment can claim

The related-work paragraph writes itself around four documented absences, each of which is an absence in a specific named paper rather than a general complaint. First, **the most complete SLAM perturbation benchmark excludes the inertial channel entirely.** Robust-SLAM spans 16 image corruptions, 4 depth corruptions, 3 trajectory perturbations and RGB-depth temporal misalignment across ~1,000 sequences and 6 SLAM systems, and includes no IMU perturbations at all; it also states explicitly that "we deliberately concentrate on investigating the performance degradation caused by individual sensor or trajectory perturbations," i.e. **no composed perturbations** (ar5iv 2402.08125). None of the autonomous-driving corruption benchmarks perturb inertial data either — Dong et al.'s alignment-level corruptions are camera-to-LiDAR, not camera-to-IMU. So a protocol that injects into both channels, and especially one that injects jointly, is positioned as filling a documented gap; **no work was found that injects visual and inertial noise simultaneously in the same run**, with Robust-SLAM's explicit non-composition as the citable contrast.

Second, **no study measures trajectory error as a function of lens-contamination coverage or opacity.** Soiling-impact numbers exist for semantic segmentation — DeepLabV3+ falling from **56.6% to 34.8% mIoU on WoodScape (−21.8 points)** and from 38.1% to 26.6% on Cityscapes (Uřičář et al., WACV 2021 / arXiv:1912.02249), with Cityscapes mIoU dropping **0.692 → 0.405 under adherent droplets** and road-marking IoU **0.594 → 0.396 with real drops and → 0.287 with synthetic ones** (Porav et al.) — and for 2D detection, where physically rendered raindrops cost only **1.18% mAP overall but 4.00% on pedestrians** (von Bernuth et al.). SLAM corruption benchmarks exist for noise, blur and weather. The intersection is empty. The detection-versus-segmentation asymmetry also predicts where sparse-feature VIO sits: nearer the detection end, since objects behind drops remain recognizable, but with the added vulnerability that VIO depends on *geometric consistency* of surviving correspondences, which the droplet-motion result says are systematically biased rather than merely sparse.

Third, **no corruption-robustness study uses EuRoC as its base dataset, and no study uses EuRoC's published IMU noise densities as the nominal anchor for a severity sweep.** Failure-or-Drift uses KITTI and 4Seasons; Robust-SLAM uses 8 Replica scenes; Robo3D, Dong et al. and RoboBEV are automotive; Zhu et al. use CERBERUS. Chen et al. use EuRoC as a base but with single operating points rather than a graded sweep. And **no work applies a graded severity sweep to the IMU channel at all** — Zhu et al.'s "nine practical perturbation settings" come closest, but whether those are graded levels or distinct configurations could not be verified.

Fourth, **the detector-level and system-level curves the design would naturally produce do not exist for ORB.** No paper plots ORB keypoint count or inlier ratio against blur kernel length; the SSMB tables cover SIFT, SuperPoint and BALF but not ORB. No paper reports ATE as a function of JPEG quality factor for ORB-SLAM or any VIO system. No paper gives an ORB- or FAST-specific repeatability-versus-quality-factor curve. No paper studies *under*-specification (assuming less noise than present) versus *over*-specification as distinct failure modes, which is the more interesting tuning question. And **no published**

ATE-versus-IMU-noise-multiplier curve was successfully extracted from any source — the field has no established "how much does ATE degrade per unit of IMU noise" relationship, which is exactly the curve the measured 1×/5×/10×/20×/50× sweep produces.

Two framing points keep these claims defensible. The honest scope statement is that synthetic corruption measures **sensitivity of an estimator to a named, parameterized stressor** — a well-posed question the literature supports strongly — and is *not* a validated proxy for field performance, which the literature supports only contestedly (see the dispute below). And the argument for synthesis over real adverse data is methodological rather than convenient: real adverse datasets confound condition with route, platform and ground-truth quality, while controlled perturbation holds geometry fixed, which is TartanAir's central defense of simulation (Wang et al., IROS 2020) and 3DCC's rationale for geometry-aware corruption over image-space filters (Kar et al., CVPR 2022). 4Seasons' repeated traversal of the same route under different conditions is the real-world analogue of a severity sweep and the best available validation companion (Wenzel et al., IJCV 2023, arXiv:2301.01147), with DIDLM the only dataset found that benchmarks visual+IMU and LiDAR+visual+IMU side by side under real adverse conditions (arXiv:2404.09622).

What is unverified, and what the literature genuinely disputes

One correction must survive into any paper built on this material, because it was introduced by an automated page summary and caught only by reading the source. **An automated summary wrongly attributed "occlusion/soiling" and "lens contamination" corruptions to "Failure or Drift? Evaluating Monocular SLAM under Synthetic and Real-World Corruptions." That paper contains no soiling, dirt, droplet or local-occlusion corruption of any kind.** Its three families are 13 image-space transforms from `imagecorruptions` (brightness, contrast, defocus blur, elastic transform, frost, Gaussian noise, impulse noise, JPEG compression, motion blur, pixelation, shot noise, snow, zoom blur), 7 geometry-aware effects from 3D Common Corruptions (far focus, near focus, flash, fog, ISO noise, low light, rain), and 3 compound effects, at 5 severities each for 115 corrupted variants of KITTI sequence 00 plus 11 clean odometry sequences — verified against the paper's full text (arXiv:2608.30690). Do not cite it for lens contamination. The closest thing to a lens-contamination operator in any corruption benchmark is the **spatter** effect, which is one of ImageNet-C's four "extra" corruptions and appears in Robust-SLAM's environmental family; **lens occlusion, soiling, droplets and dirt are not separate categories in any corruption benchmark found**.

The genuine open dispute concerns whether synthetic corruption robustness transfers to real degradation, and it should be presented as unresolved. **Taori et al. evaluated 196 ImageNet models across 211 test conditions and found "little to no transfer of robustness from current synthetic to natural distribution shift," with effective robustness on the two not correlating** (NeurIPS 2020, arXiv:2007.00644). **Hendrycks et al. report the opposite — that synthetic corruptions *do* correlate with corruptions appearing in the wild, "contradicting speculations from previous work"** (ICCV 2021, arXiv:2006.16241). Mintun et al. partially reconcile them by showing the transfer depends on perceptual similarity between the injected and real corruption, but do not settle the headline question. A paper citing either side as settled is attackable; citing the dispute and then claiming only estimator-sensitivity is not.

The remaining caveats are verification status rather than disagreement, and they are listed here so the paper's bibliography can be checked before submission rather than after review.

Claim	Status	Action before publication
All ADIS16448 datasheet figures (noise density, ARW, bias stability, misalignment, linear-g)	From a **third-party aggregator** (radiolocman.com), not analog.com; both official ADI PDF URLs failed	Verify against the official ADI datasheet (Rev. H)
ADIS16448 **accelerometer** noise density and velocity random walk	**Not verified at all** — the aggregator table omits those rows	Cite ADI directly, or use EuRoC's 2.0e−3 m/s²/√Hz as the operative value and say so
ADIS16448 noise density (2.356e−4) vs ARW (1.920e−4) differing by 1.23×	Real internal inconsistency in the published specs; ADI likely specifies them at different filter/decimation settings	Report the discrepancy rather than picking one
ADIS16448 linear-g sensitivity 0.015 °/s/g	Some secondary summaries quote 0.009 °/s/g for this family; unresolved	Resolve from the official datasheet
EuRoC's four published noise values	Taken from the **OpenVINS republication**, cross-checked against ORB-SLAM3's EuRoC.yaml; the primary `mav0/imu0/sensor.yaml` was not fetched	Cite the dataset's own `sensor.yaml`
The 2.512× OpenVINS divergence threshold and the accelerometer-dominance attribution	**Search-engine summary only.** A WebFetch of arXiv:2609.18628 read only uncompressed metadata; the ATE tables, multiplier range and system list were in compressed streams	Re-read the PDF before quoting any number
Chen et al., "Selective Sensor Fusion for Neural Visual-Inertial Odometry" venue	**Unverified.** Widely cited as CVPR 2019; the ar5iv rendering shows no venue	Verify before citing a venue; its IMU noise magnitudes are also unquantified in the retrieved text
ImageNet-C per-severity constants	Read from a **third-party mirror** of `imagenet_c`	Pull from `github.com/hendrycks/robustness`
EMVA 1288 equations	Quoted from the Standard Group chair's handout and compliant vendor datasheets; the **normative PDF could not be downloaded**	Cite EMVA Standard 1288, Release 4.0 Linear, 21 June 2021 directly
Allan-variance read-off constants √3 (quantization) and 1/√2 (rate ramp)	**Not verified**; only the 0.664 bias-instability constant and the $\tau=1/\tau=3$ read-off points were confirmed	Check against El-Sheimy et al. 2008 or IEEE Std 952-2020
Gyro ARW reported in "rad/s" and accel bias instability in "m/s²" in the Sensors 2024 MEMS-grade table	**Dimensionally inconsistent as printed** (DOI 10.3390/s24061953); most likely rad/s/√Hz	Flag the units ambiguity rather than silently fixing it
TUM-VI IMU parameters	Paper omits the `/√Hz` on all four; must be read as densities (ar5iv 1804.06120)	Note the omission when citing

Claim	Status	Action before publication
VINS-Mono `acc_n` / `gyr_n` units	Config comments give no units; community debate exists about the dt scaling inside its preintegration	Do not assert the units without reading `integration_base.h`
"Photometric calibration reduces ORB-SLAM performance"	**Search-summary only**; the arXiv PDF for 1705.04300 would not parse	Fetch the RA-L version or state qualitatively
Ehsan et al. JPEG figures (SIFT to 50%, SURF to 95%)	**Search-summary only**, and expressed as compression ratio not quality factor	Do not convert to Q; verify or drop
ORB-SLAM3 rolling-shutter "decimeter vs centimeter" figure	**Student thesis, record summary only** (DiVA 1768623)	Treat as non-peer-reviewed
Robust-SLAM's ORB-SLAM3 ATE cells (clean 0.0145 m, Gaussian noise 0.535 m, motion blur → failure)	From a single-pass HTML read of a very long page	Re-verify specific cells before quoting as headline numbers
Liu, Tanaka & Okutomi IEEE TIP 2013 volume/pages/DOI	**Unverified** (commonly cited as 22(12):5226–5237, doi:10.1109/TIP.2013.2283400)	Verify
DOIs for ORB-SLAM3, OKVIS, VINS-Mono, OpenVINS, Kalibr (IROS 2013), Nikolic et al. ICRA 2014	**Not verified this session**	Verify
MT9V034 electron-level quantities (full well, conversion gain, read noise in e^-, dark current, DSNU/PRNU)	**None are published.** All derived numbers in this report are brackets from comparable sensors	Present as brackets or measure from dark/flat frames
EuRoC exposure times	**Not found in any source**	Read from the bags or state 1–20 ms as an assumption with the TUM mono 0.05–20 ms precedent
EuRoC cam0 intrinsics (fx ≈ 458.65, fy ≈ 457.30, cx ≈ 367.22, cy ≈ 248.38)	**From training knowledge, not verified**	Read from the dataset's `sensor.yaml` before any blur-magnitude calculation
ObjectNet, ImageNet-A, ImageNet-Sketch, ImageNet-V2, ACDC	**Not researched or verified**	Do not attribute specifics; ACDC is believed to be a segmentation dataset, not a trajectory benchmark
An ICCV/CVPR workshop challenge on soiling detection	**Could not be confirmed to exist**; WoodScape-based challenges found were for fisheye object detection	Do not assert a soiling challenge existed

Claim	Status	Action before publication
Hahner et al. LiDAR-fog simulation (underlying Robo3D's fog model); RIGSEC's original droplet citation; Sadeghi et al. environment-map integration; Hedborg et al. RS bundle adjustment; Barreda Pupo thesis	**Not retrieved**	Verify before citing
Plötz & Roth (DND), Abdelhamed et al. (SIDD), Lehtinen et al. (Noise2Noise), Mäkitalo & Foi, Hasinoff et al., Woodman UCAM-CL-TR-696, Trawny & Roumeliotis TR 2005-002	**Referenced from memory, not retrieved**	Verify
arXiv IDs 2609.18628, 2603.17165, 2608.30690, 2608.27181, 2601.04185, 2605.09383, 2603.26740, 2512.08653, 2607.13405, 2607.22123	Several postdate reliable knowledge; only some were confirmed to return real PDFs with the stated titles	Re-verify IDs and titles

Two softer gaps deserve a sentence in a limitations paragraph. **No peer-reviewed paper states the config-inflation practice as such** — the evidence is the configs themselves plus Kalibr's "optimistic" caveat — so the re-balancing argument should be made from primary config files rather than attributed to a paper. And **no measured defect density exists for a 6 μm global-shutter machine-vision sensor**, nor any specification of allowed blemish counts for the MT9V034; the Chapman formulas come from consumer DSLR studies parameterized by ISO, so mapping them to machine-vision analog gain is approximate and should be presented as an order-of-magnitude statement.

Conclusion

The sharpest thing this material establishes is that for this hardware the physically defensible envelope is *narrow and low*, and that narrowness is the contribution rather than a limitation. A well-exposed mono8 frame from a 6 μm global-shutter sensor carries about 1.4 DN of total noise at mid-grey; a MEMS IMU whose datasheet gyro noise density matches a consumer Bosch part but whose bias specification does not; a sequence whose own easy-grade designation means blur and illumination stress are near zero outside the initialization segment. Against that envelope, most of the corruption literature is either inapplicable (weather, rolling shutter, demosaicing) or calibrated for a different regime (ImageNet-C constants tuned for 224 px crops, denoising σ values 5–20× above the sensor's physics, automotive soiling geometry defined on a windshield 30 cm from the lens). The experiment's defensibility therefore comes from

rejection and from unit discipline — physical units at every severity axis, EMVA 1288 and Allan variance as the anchors, and an explicit list of the corruptions the hardware cannot produce — rather than from breadth.

The second insight is that the measured nondeterminism floor, not the literature, is the binding constraint on what can be claimed. A 3.4 mm pooled standard deviation on a 40–44 mm baseline means roughly 6 mm of resolving power at five seeds per condition and 3 mm at twenty, which is why the 5× result reads as an improvement and is not one, and why the degradation knee at 20× is believable at 7.6σ. That number reframes the inertial result from a null finding into a bounded one: ORB-SLAM3 is demonstrably insensitive to white-noise inflation across the entire 4–40× band that OKVIS and VINS-Mono already ship, which means the interesting axes are elsewhere — the bias random walk that every production config *deflates* by 5–75× and that no published study sweeps, and the deterministic scale, misalignment and g-sensitivity terms that the ADIS16448 datasheet quantifies and the standard four-parameter model structurally cannot absorb. Those are the corruptions where this sensor suite licenses something the field has not measured, and where a feature-based optimizer's response is genuinely unknown rather than merely unpublished.